

Next meeting will be on Sunday(19-07-2026)

Week 3 — Conformal Calibration, Robustness & Feature-Selection Results (Days 15–18)

With frozen data and first prototypes in hand, Team A builds the statistical guarantee (Algorithm 2) and the coverage-verification harness, Team B builds the noise channel, Lipschitz estimate and certified radius that validate Proposition 2, and Team C completes the Article-2 benchmark and starts writing it up.

Team A — Conformal Calibration (Algorithm 2), Coverage Harness & Statistics

Members: Interns 1, 2, 3 (Amon Koike, Mohammed Owais, Iwo Wojtakajtis)

Focus this week: Own the paper's statistical-guarantee centrepiece: implement split-conformal calibration on real prototypes and build the coverage-verification pipeline.

Day	Tasks	Deliverables
Day 15	Implement Algorithm 2 (CQ-ZDR): nonconformity score $s = 1 - \max_c F(p(x), p_c)$; threshold $q = s_{(k)}$, $k = \lceil (1-\alpha)(n+1) \rceil$. Test Q on Team B's real prototypes for dataset 1 at $\alpha = 0.05$.	Conformal calibration module First threshold q (dataset 1)
Day 16	Build the coverage-verification harness: empirical false-zero-day rate on a held-out KNOWN set must be $\leq \alpha$. Add α -sweep (0.01–0.20) to produce the headline coverage curve later.	Coverage-verification harness α -sweep scaffold
Day 17	Integrate the 5-seed statistical protocol: mean $\pm$ std, 95% CI, paired t-test, McNemar, Holm–Bonferroni, Cohen's d. Dry-run significance tests on baseline-vs-baseline as a sanity check.	Statistics pipeline (complete) Significance dry-run report
Day 18	Extend conformal calibration + coverage to all three datasets as prototypes arrive. Log achieved false-zero-day rate vs target α per dataset.	Conformal thresholds (all datasets) Coverage table v1

Team B — Noise Channel, Lipschitz Estimate, Certified Radius & Proposition-2 Quantities

Members: Interns 4, 5 (Arjun E Naik, La Wun Nannda)

Focus this week: Build the robustness and theory-validation machinery: depolarizing noise, encoder Lipschitz constant, certified radius (Algorithm 3, line 8), and the empirical quantities behind Proposition 2.

Day	Tasks	Deliverables
Day 15	Implement the depolarizing channel Λ_p and confirm Proposition 1 numerically: $D_{\text{tr}}(\Lambda_p(\rho), \Lambda_p(\sigma)) = (1-p) \cdot D_{\text{tr}}(\rho, \sigma)$. Expose p as both a NISQ-noise model and a regulariser knob.	Depolarizing channel Prop-1 numerical check
Day 16	Estimate the encoder Lipschitz constant L_ϕ per dataset (95th-percentile of trace-distance / input-distance over sampled pairs). Report L_ϕ with its distribution.	L_ϕ estimator Per-dataset L_ϕ values
Day 17	Implement the certified radius $R = m / (2(1-p) \cdot L_\phi \cdot C_f)$ (Algorithm 3, line 8). Produce the certified-radius distribution on correctly-classified test samples.	Certified-radius module Radius distribution (dataset 1)

Day	Tasks	Deliverables
Day 18	Measure Proposition-2 quantities: F_{in} (fidelity-to-own-prototype) and F_{out} (max fidelity for zero-day). Compute the predicted separable-budget gap $\epsilon^* = (g(F_{in}) - g(F_{out})) / (2(1-p)L_\phi)$.	F_{in} / F_{out} measurements Predicted ϵ^* per dataset

Team C — MO-AQPSO Full Benchmark, Feature Hand-off & Article-2 Drafting

Members: Interns 6, 7 (Kevin Wong, Maral Mahmoudi Kamelabad)

Focus this week: Complete the comparative study, deliver the final feature subsets into QS-Net, and begin writing Article 2 so results and text advance together.

Day	Tasks	Deliverables
Day 15	Finalise the five-selector benchmark with 5-seed runs and 95% CIs (reuse Team A statistics helpers). Add stability (Jaccard across seeds) and redundancy metrics.	Final selector benchmark (5-seed) Stability/redundancy metrics
Day 16	Freeze the FINAL Top-10 feature subset per dataset from MO-AQPSO. Deliver frozen subsets to Team B for QS-Net training (links Article 2 → Article 1).	FROZEN feature subsets Hand-off note to Team B
Day 17	Set up the Article-2 experiment: selector → Zero-Day AUROC (MI / RF / BPSO / QPSO / MO-AQPSO). Run the first version on dataset 1.	Zero-Day-AUROC-by-selector table (v1)
Day 18	Extend selector→zero-day-AUROC to all datasets. Draft Article-2 Introduction and Related Work.	All-dataset selector→AUROC results Article-2 Intro + Related Work draft

★ **MILESTONE** — Day 15 — Conformal calibration runs on real prototypes (dataset 1); Team C delivers frozen feature subsets to Team B.